

Alan Zhang

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Software Engineer at a Fortune 500 co. building LLM-powered tools on top of serverless AWS data pipelines. (U.S. Citizen)

Work Experience

Toll Brothers

Software Engineer

May 2025 – Present, 1 year 3 months

Fort Washington, PA

America's largest luxury homebuilder, a Fortune 500 company (\$10.8B+ in annual revenue), delivering 11K+homes/year

- Shipped an LLM-powered QA platform automating compliance analysis for land survey submissions, saving **2 hours per submission** across **5–10** daily nationwide reviews.
- Built a CI/CD-integrated code review pipeline (Claude Sonnet) that reconstructs full file context from unified diffs and posts inline comments, cutting review time by **30 minutes per PR**.
- Automated PDF batch ingestion with a Node.js CLI orchestrating uploads across CMS, S3, and Workfront APIs, replacing a week-long manual process and saving **100+ hours per batch**.
- Built a reusable server-side “**Print View**” document generation system producing dynamic community brochures with listing data, QR codes, and configurable filtering for **thousands of monthly users**.
- Architected a serverless AWS data pipeline (Lambda, Glue, Athena, EventBridge, CDK) aggregating data from **4 competitors'** undocumented APIs into a unified SQL-queryable warehouse requiring 0 operational maintenance.
- Stack:** TypeScript, Python, Node.js, AWS (Lambda, S3, DynamoDB, Bedrock), Next.js

Personal Projects

YSC Lunch Soccer | *Next.js, React Native, Express.js, Supabase, PostgreSQL, Redis, BullMQ* [Live Site](#) | [GitHub](#)

- Built a production full-stack scheduling platform replacing group-chat coordination for a **200-member** soccer community with, attendance tracking, role-based administration, and Stripe payments.
- Implemented real-time attendance updates with Supabase Realtime (Postgres CDC over WebSocket), Redis caching, and endpoint-tiered rate limiting.
- Decoupled notification delivery from the request path with a BullMQ + Redis job queue, processing Twilio SMS and transactional emails asynchronously with **3-attempt exponential backoff** for fault tolerance.
- Engineered a skill-based team balancing algorithm using **6 tiers** and a **snake draft** to auto-distribute randomized players across dynamically sized teams, eliminating **~15 minutes** of manual setup per session.
- Designed a layered authentication and authorization system using Supabase JWTs, **httpOnly** cookies, **silent token refresh**, and middleware-based request validation for secure route-level access control.

ApplyLog | *Chrome Extension (MV3), TypeScript, Next.js, Supabase, Gmail API, OAuth 2.0* [Live Site](#) | [GitHub](#)

- Built a Chrome Extension supporting Greenhouse, Lever, Ashby, and Workday through a unified scraping abstraction that synchronizes **500+ applications** into Supabase.
- Implemented PKCE OAuth 2.0 authentication using `chrome.identity` with concurrent token refresh deduplication to eliminate redundant authentication flows.
- Designed an event-driven matching pipeline that embeds scraped job descriptions with OpenAI, computes cosine similarity against the user's resume embedding, and stores match scores automatically.
- Migrated Gmail synchronization from client-side polling to scheduled server-side jobs, reducing worst-case status staleness from indefinite to **under 15 minutes** and propagating updates live with Supabase Realtime.

Education

University of Maryland – College Park

Bachelor of Science in Computer Science

Aug. 2022 – May 2025

College Park, MD

Technical Skills

Programming Languages: JavaScript, TypeScript, Python, Java

Front-end/Back-end/Mobile Technologies: React.js, React Native, Next.js, Node.js, HTML, CSS, SCSS, Tailwind CSS, Flask, Express.js, GraphQL, REST APIs, Stripe, Twilio, Expo, Expo Router

Database & ORM Technologies: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, Prisma, SQL

Cloud & DevOps: AWS Amplify, Cognito, S3, CloudFront, Git, GitHub, Bitbucket, Docker, Jira, Supabase